

Objective

The goal of EDA is not just to make charts.

Its purpose is to answer business questions and understand the data before building any ML model.

At the end of this phase, we should know:

What kind of customers open CASA accounts? Which branches perform best? Which lead source gives the highest conversion? Which variables influence conversion? Is the data balanced? Are there missing values or outliers? Which features are important?

```
In [1]: import pandas as pd

customers = pd.read_csv("DATASETS OF CASA PRO/customers.csv")
branches = pd.read_csv("DATASETS OF CASA PRO/branches.csv")
leads = pd.read_csv("DATASETS OF CASA PRO/leads.csv")
accounts = pd.read_csv("DATASETS OF CASA PRO/casa_accounts.csv")
transactions = pd.read_csv("DATASETS OF CASA PRO/transactions.csv")
rm = pd.read_csv("DATASETS OF CASA PRO/relationship_managers.csv")
sales = pd.read_csv("DATASETS OF CASA PRO/sales_targets.csv")
campaign = pd.read_csv("DATASETS OF CASA PRO/campaign_history.csv")
```

1)CUSTOMER DATA EXPLORATIONS

Chunk 1 — Customer Data Exploration

Objective: Understand the customer profile and check whether the customer dataset is suitable for further analysis.

We look at:

Customer demographics Numerical distributions Missing values Duplicate records Basic statistics Potential unusual values

```
In [2]: customers.head(20)
```

Out[2]:

	customer_id	age	gender	occupation	annual_income	salary_account	credit_score	home_branch
0	C00001	25	Male	Teacher	72834	No	697	B005
1	C00002	56	Male	Business	44663	Yes	810	B001
2	C00003	41	Female	Business	47350	Yes	833	B010
3	C00004	41	Male	Student	133712	No	636	B005
4	C00005	29	Male	Business	183245	No	829	B010
5	C00006	46	Female	Teacher	66262	No	623	B001
6	C00007	28	Male	Teacher	140666	Yes	698	B007
7	C00008	59	Female	Business	175024	Yes	774	B002
8	C00009	55	Female	Business	59242	No	711	B003
9	C00010	50	Male	Business	161225	Yes	729	B002
10	C00011	40	Male	Teacher	128019	Yes	738	B009
11	C00012	21	Female	Student	35105	Yes	837	B006
12	C00013	40	Male	Engineer	93143	Yes	621	B008
13	C00014	25	Male	Teacher	154592	Yes	667	B009
14	C00015	59	Female	Teacher	171373	Yes	782	B005
15	C00016	46	Female	Student	165679	No	630	B004
16	C00017	35	Male	Business	35514	Yes	750	B004
17	C00018	21	Male	Engineer	90015	Yes	831	B001
18	C00019	42	Male	Teacher	103001	No	654	B009
19	C00020	29	Female	Teacher	153987	No	648	B002

```
In [3]: customers.columns
```

```
Out[3]: Index(['customer_id', 'age', 'gender', 'occupation', 'annual_income',
              'salary_account', 'credit_score', 'home_branch'],
              dtype='object')
```

```
In [4]: customers.isnull().sum()
```

Out[4]: customer_id 0
age 0
gender 0
occupation 0
annual_income 0
salary_account 0
credit_score 0
home_branch 0
dtype: int64

In [5]: customers.duplicated().sum()

Out[5]: 0

In [6]: customers.describe()

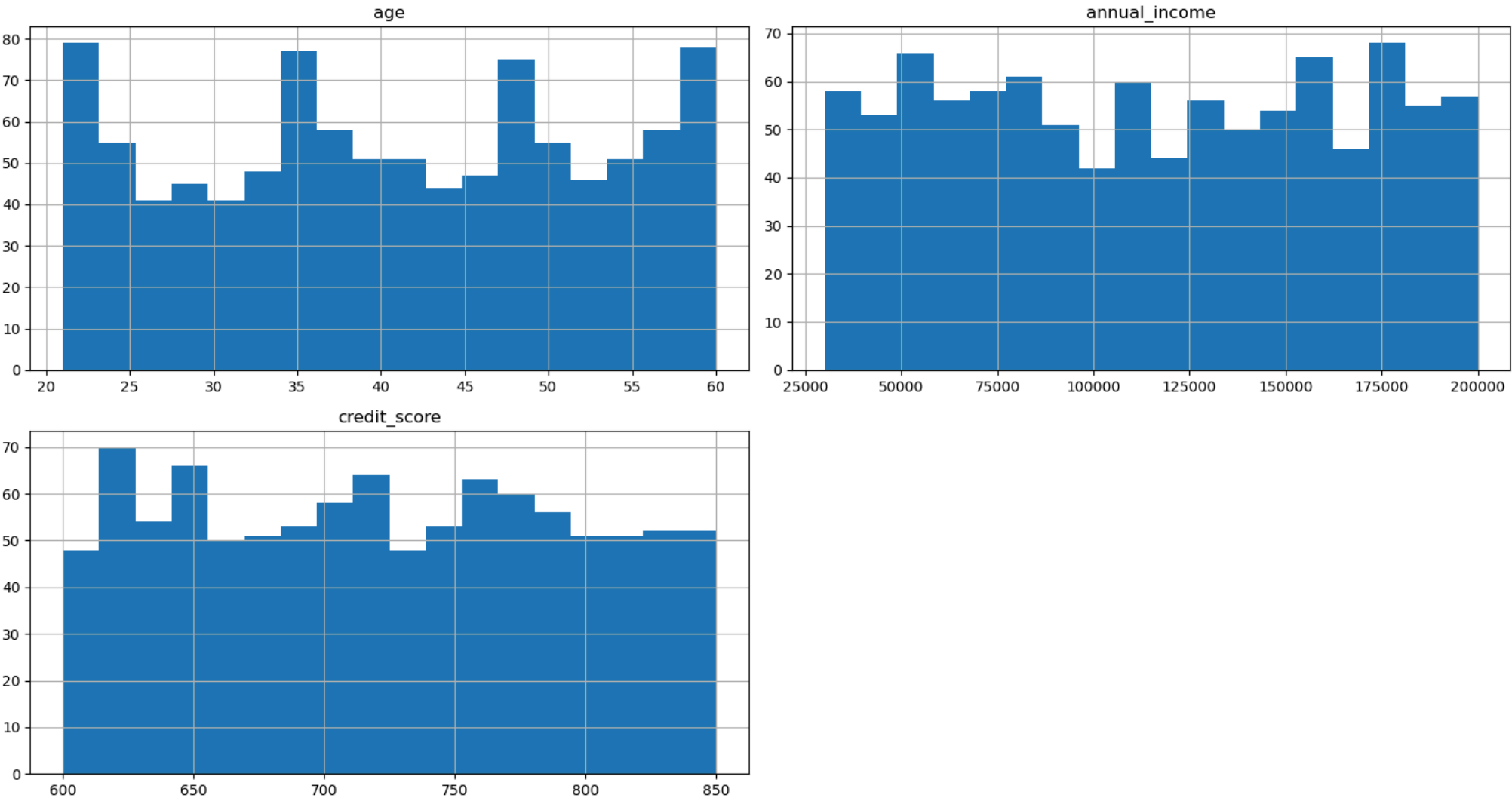
Out[6]:

	age	annual_income	credit_score
count	1000.000000	1000.000000	1000.000000
mean	40.796000	114953.698000	723.178000
std	11.624953	49978.549869	71.981189
min	21.000000	30075.000000	600.000000
25%	31.000000	70865.000000	659.000000
50%	41.000000	114663.000000	721.000000
75%	51.000000	158031.500000	783.000000
max	60.000000	199989.000000	850.000000

In [7]: customers.shape

Out[7]: (1000, 8)

In [8]: import matplotlib.pyplot as plt
customers.hist(figsize= (15,8), bins = 18),
plt.ylabel("Number of Customers"),
plt.tight_layout()
plt.show()



2. BRANCH DATA EXPLORATION

Chunk 2 — Branch Data Exploration

Objective: Understand the branch structure and the distribution of monthly targets across branches.

You currently analyze:

Branch ID Branch name City Monthly target

In [9]: branches.head(20)

Out[9]:

	branch_id	branch_name	city	state	monthly_target
0	B001	Branch 1	Bengaluru	KA	679
1	B002	Branch 2	Pune	KA	414
2	B003	Branch 3	Mumbai	TS	352
3	B004	Branch 4	Bengaluru	TS	516
4	B005	Branch 5	Bengaluru	KA	347
5	B006	Branch 6	Mumbai	KA	558
6	B007	Branch 7	Bengaluru	TS	401
7	B008	Branch 8	Hyderabad	KA	529
8	B009	Branch 9	Pune	KA	688
9	B010	Branch 10	Mumbai	TS	516

In [10]:

```
branches = branches.drop("state",axis=1)
```

In [11]:

```
branches.columns
```

Out[11]:

Index(['branch_id', 'branch_name', 'city', 'monthly_target'], dtype='object')

In [12]:

```
branches.isnull().sum()
```

Out[12]:

branch_id 0
branch_name 0
city 0
monthly_target 0
dtype: int64

In [13]:

```
branches.duplicated().sum()
```

Out[13]:

0

In [14]:

```
branches.describe()
```

Out[14]:

	monthly_target
count	10.00000
mean	500.00000
std	122.29836
min	347.00000
25%	404.25000
50%	516.00000
75%	550.75000
max	688.00000

In [15]:

```
import matplotlib.pyplot as plt

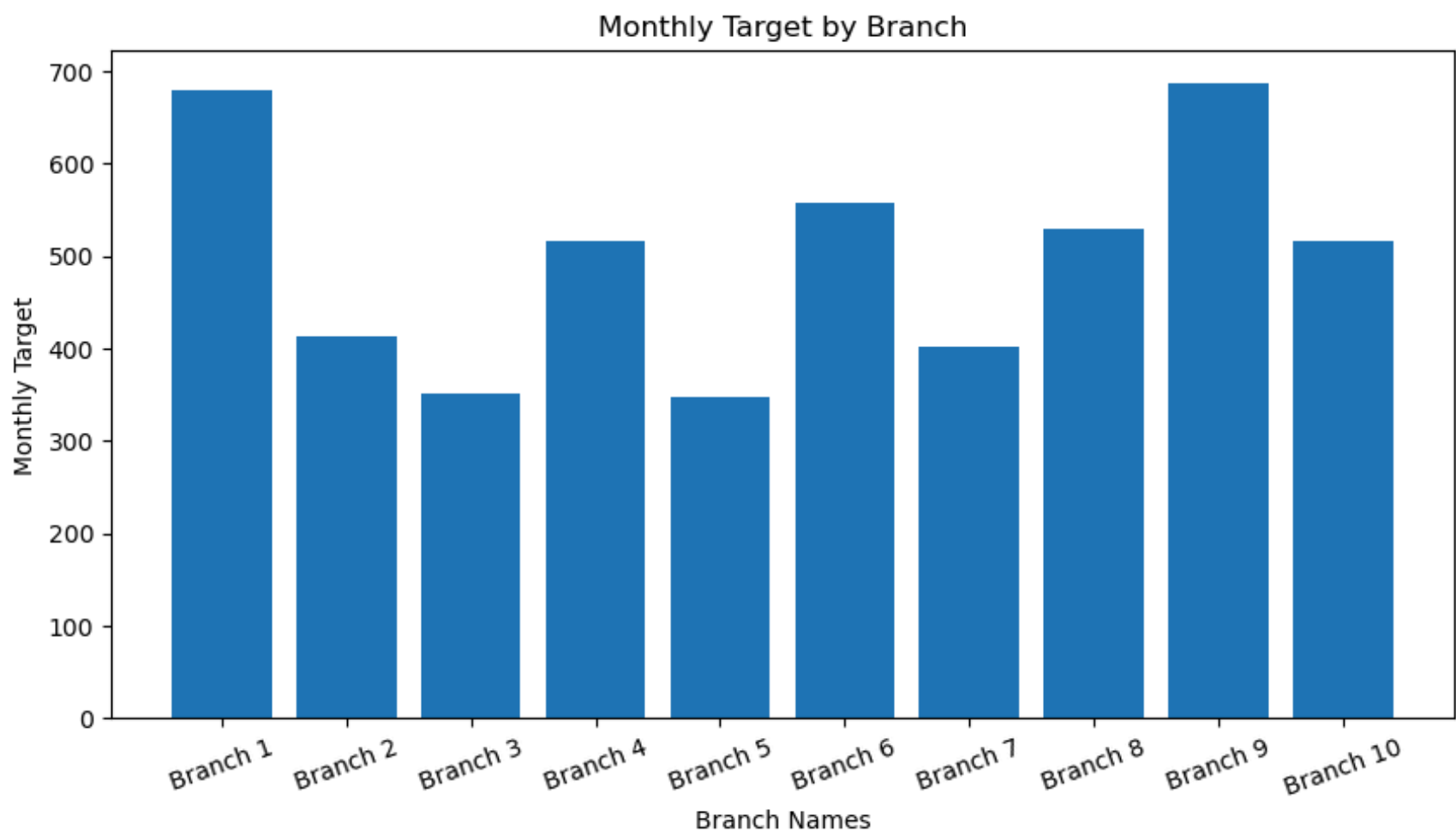
plt.figure(figsize=(10,5))

plt.bar(branches["branch_name"], branches["monthly_target"])

plt.title("Monthly Target by Branch")
plt.xlabel("Branch Names")
plt.ylabel("Monthly Target")

plt.xticks(rotation=20)

plt.show()
```



3)Leads DATA EXPLORATION

Objective: Understand the CASA lead population and identify factors associated with successful conversion.

Main variables:

Lead source RM Converted / not converted

This is currently one of the most valuable sections because converted is key outcome variable.

```
In [16]: leads.columns
```

Out[16]: Index(['lead_id', 'customer_id', 'rm_id', 'source', 'converted'], dtype='object')

```
In [17]: leads.head(20)
```

Out[17]:

	lead_id	customer_id	rm_id	source	converted
0	L00001	C00001	RM016	Walk-in	0
1	L00002	C00002	RM028	Digital	1
2	L00003	C00003	RM005	Digital	0
3	L00004	C00004	RM021	Digital	1
4	L00005	C00005	RM011	Digital	1
5	L00006	C00006	RM014	Digital	0
6	L00007	C00007	RM008	Referral	1
7	L00008	C00008	RM012	Referral	1
8	L00009	C00009	RM007	Referral	1
9	L00010	C00010	RM017	Walk-in	0
10	L00011	C00011	RM010	Walk-in	1
11	L00012	C00012	RM012	Digital	0
12	L00013	C00013	RM003	Walk-in	1
13	L00014	C00014	RM016	Digital	0
14	L00015	C00015	RM004	Referral	1
15	L00016	C00016	RM016	Digital	1
16	L00017	C00017	RM022	Campaign	0
17	L00018	C00018	RM028	Campaign	1
18	L00019	C00019	RM022	Walk-in	1
19	L00020	C00020	RM013	Walk-in	1

```
In [18]: leads.describe()
```

Out[18]:

	converted
count	1000.000000
mean	0.626000
std	0.484106
min	0.000000
25%	0.000000
50%	1.000000
75%	1.000000
max	1.000000

In [19]:

```
#no of CASA account where Lead converted succesfully
leads["converted"].value_counts()
```

Out[19]:

converted	
1	626
0	374
Name: count, dtype: int64	

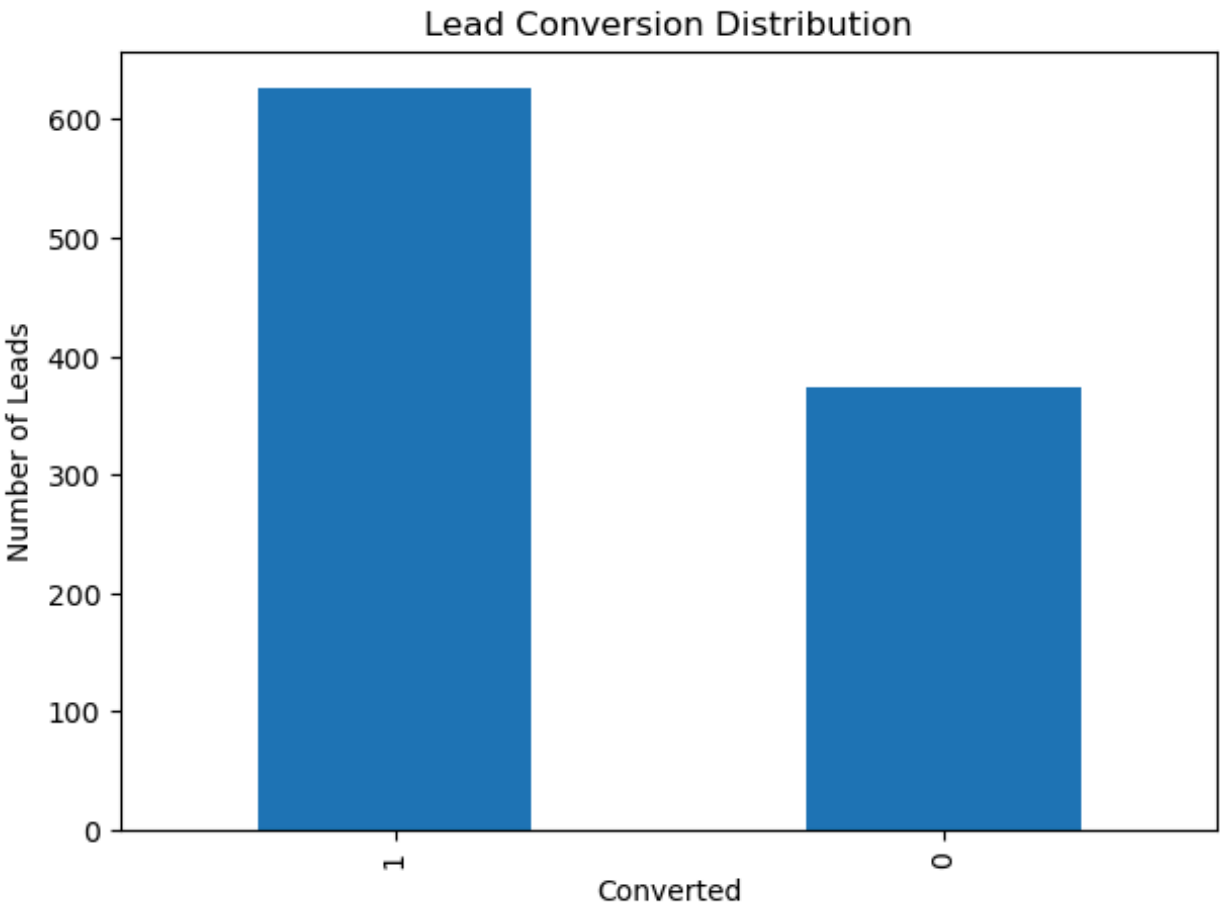
In [20]:

```
plt.figure(figsize=(7,5))

leads["converted"].value_counts().plot(kind="bar")

plt.title("Lead Conversion Distribution")
plt.xlabel("Converted")
plt.ylabel("Number of Leads")

plt.show()
```



In [21]:

```
source_analysis = leads.groupby("source").agg(
    total_leads=("lead_id", "count"),
    converted_leads=("converted", "sum"),
    conversion_rate=("converted", "mean")
)

source_analysis["conversion_rate"] = (
    source_analysis["conversion_rate"] * 100
)

source_analysis = source_analysis.sort_values(
    "conversion_rate",
    ascending=False
)

print(source_analysis)
```

	total_leads	converted_leads	conversion_rate
source			
Referral	261	170	65.134100
Campaign	244	155	63.524590
Digital	233	146	62.660944
Walk-in	262	155	59.160305

In [22]:

```
# RM Lead Volume and Conversion Analysis
```

```
rm_analysis = leads.groupby("rm_id").agg(
```

```
total_leads=("lead_id", "count"),
converted_leads=("converted", "sum"),
conversion_rate=("converted", "mean")
)

# Convert conversion rate to percentage
rm_analysis["conversion_rate"] = (
    rm_analysis["conversion_rate"] * 100
)

# Sort by conversion rate
rm_analysis = rm_analysis.sort_values(
    "conversion_rate",
    ascending=False
)

print(rm_analysis)
```

	total_leads	converted_leads	conversion_rate
rm_id			
RM013	41	32	78.048780
RM007	31	24	77.419355
RM012	41	31	75.609756
RM005	24	17	70.833333
RM030	23	16	69.565217
RM006	36	25	69.444444
RM011	38	26	68.421053
RM020	36	24	66.666667
RM016	41	27	65.853659
RM002	32	21	65.625000
RM004	29	19	65.517241
RM023	20	13	65.000000
RM001	31	20	64.516129
RM024	39	25	64.102564
RM021	36	23	63.888889
RM010	33	21	63.636364
RM003	22	14	63.636364
RM027	41	26	63.414634
RM008	35	22	62.857143
RM022	36	22	61.111111
RM025	36	22	61.111111
RM019	34	20	58.823529
RM017	31	18	58.064516
RM018	36	20	55.555556
RM015	27	15	55.555556
RM026	36	20	55.555556
RM028	42	22	52.380952
RM009	34	16	47.058824
RM029	23	10	43.478261
RM014	36	15	41.666667

highest lead is genrated through referral. Referral is the best source

Referral leads achieved the highest CASA conversion rate at 65.13%, followed by Campaign (63.52%) and Digital (62.66%), while Walk-in leads had the lowest conversion rate at 59.16%. This suggests that referral-based acquisition is currently the most effective lead source for CASA conversion.

Relational manager coversion analysis Top: RM013 → 78.05% Bottom: RM014 → 41.67% Difference → approximately 36.38 percentage points

4)Accounts data exploration

Chunk 4 — CASA Account Exploration

Objective: Understand the characteristics of successfully opened CASA accounts and check the quality of the account data.

To check:

Account structure Missing values Descriptive statistics Duplicates

This becomes useful later when connecting leads → customers → CASA accounts.

In [23]:

accounts.head(20)

Out[23]:

	account_id	customer_id	branch_id	account_type	current_balance
0	A00001	C00002	B001	Savings	25458
1	A00002	C00004	B005	Savings	35708
2	A00003	C00005	B010	Current	36023
3	A00004	C00007	B007	Current	28351
4	A00005	C00008	B002	Current	12285
5	A00006	C00009	B003	Savings	6094
6	A00007	C00011	B009	Savings	39509
7	A00008	C00013	B008	Savings	47911
8	A00009	C00015	B005	Savings	15042
9	A00010	C00016	B004	Current	6390
10	A00011	C00018	B001	Current	45507
11	A00012	C00019	B009	Savings	27255
12	A00013	C00020	B002	Current	19252
13	A00014	C00021	B001	Current	33069
14	A00015	C00022	B004	Current	11009
15	A00016	C00025	B008	Current	12591
16	A00017	C00026	B007	Savings	32189
17	A00018	C00028	B006	Current	41970
18	A00019	C00029	B009	Savings	39832
19	A00020	C00030	B004	Savings	18473

In [24]:

accounts.isnull().sum()

Out[24]:

account_id0
customer_id0
branch_id0
account_type0
current_balance0
dtype: int64

In [25]:

accounts.describe()

Out[25]:

	current_balance
count	626.000000
mean	27124.060703
std	14359.747460
min	1214.000000
25%	14706.750000
50%	28377.500000
75%	39904.750000
max	49996.000000

In [26]:

accounts.duplicated().sum()

Out[26]:

0

5)Transactions data exploration

Chunk 5 — Transaction Exploration

Objective: Understand customer transaction behavior and identify whether transaction activity could be useful for CASA analysis.

To examine:

Transaction amounts Distribution Descriptive statistics Missing values

This becomes particularly important for later question:

Is higher customer activity associated with CASA conversion?

In [27]:

transactions.head(20)

Out[27]:

	txn_id	account_id	txn_type	amount
0	T000001	A00001	Debit	2768
1	T000002	A00001	Credit	2494
2	T000003	A00001	Debit	7653
3	T000004	A00002	Credit	3102
4	T000005	A00002	Credit	9910
5	T000006	A00002	Credit	9008
6	T000007	A00003	Credit	311
7	T000008	A00003	Credit	8589
8	T000009	A00003	Debit	6211
9	T000010	A00004	Credit	2009
10	T000011	A00004	Credit	384
11	T000012	A00004	Debit	5805
12	T000013	A00005	Credit	2043
13	T000014	A00005	Credit	2208
14	T000015	A00005	Credit	3812
15	T000016	A00006	Debit	7165
16	T000017	A00006	Credit	3317
17	T000018	A00006	Credit	637
18	T000019	A00007	Credit	5859
19	T000020	A00007	Credit	1587

In [28]:

transactions.isnull().sum()

Out[28]:

txn_id0
account_id0
txn_type0
amount0
dtype: int64

In [29]:

transactions.describe()

Out[29]:

	amount
count	1878.000000
mean	5075.757721
std	2873.560225
min	100.000000
25%	2559.250000
50%	5163.500000
75%	7533.000000
max	9996.000000

6)Relational manager work data exploration

Chunk 6 — RM Exploration

Objective: Understand Relationship Manager characteristics and investigate whether RM-level factors are associated with CASA conversion.

To examine:

RM experience Monthly target Lead volume Conversion rate

We also combined RM and lead data to compare experience with conversion.

In [30]:

rm.head(20)

Out[30]:

	rm_id	branch_id	experience_years	monthly_target
0	RM001	B001	6	37
1	RM002	B002	3	33
2	RM003	B003	13	41
3	RM004	B004	2	25
4	RM005	B005	7	26
5	RM006	B006	6	74
6	RM007	B007	6	58
7	RM008	B008	5	71
8	RM009	B009	1	66
9	RM010	B010	8	54
10	RM011	B001	2	79
11	RM012	B002	7	25
12	RM013	B003	9	38
13	RM014	B004	14	60
14	RM015	B005	10	76
15	RM016	B006	14	43
16	RM017	B007	10	32
17	RM018	B008	12	24
18	RM019	B009	1	62
19	RM020	B010	4	69

In [31]:

rm.isnull().sum()

Out[31]:

rm_id0
branch_id0
experience_years0
monthly_target0
dtype: int64

In [32]:

rm.duplicated()

Out[32]:

0False
1False
2False
3False
4False
5False
6False
7False
8False
9False
10False
11False
12False
13False
14False
15False
16False
17False
18False
19False
20False
21False
22False
23False
24False
25False
26False
27False
28False
29False
dtype: bool

In [33]:

rm.describe()

Out[33]:

	experience_years	monthly_target
count	30.000000	30.000000
mean	7.800000	47.000000
std	4.262102	18.463384
min	1.000000	24.000000
25%	5.000000	33.000000
50%	7.000000	42.000000
75%	11.750000	61.500000
max	14.000000	79.000000

In [34]:

```
rm.corr(numeric_only = True)
```

Out[34]:

	experience_years	monthly_target
experience_years	1.000000	-0.199817
monthly_target	-0.199817	1.000000

In [35]:

```
import matplotlib.pyplot as plt

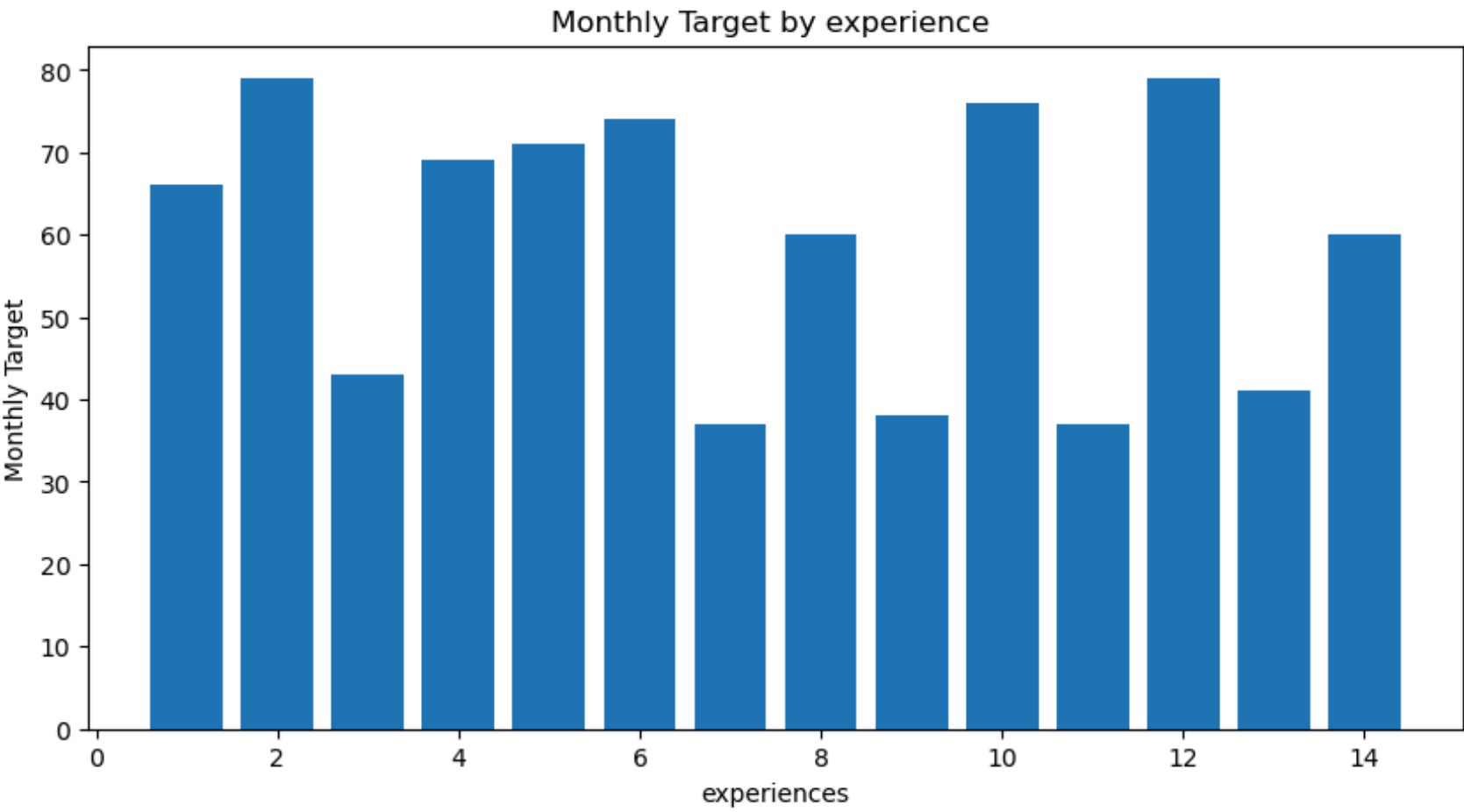
plt.figure(figsize=(10,5))

plt.bar(rm["experience_years"], rm["monthly_target"])

plt.title("Monthly Target by experience ")
plt.xlabel("experiences")
plt.ylabel("Monthly Target")

plt.xticks(rotation=0)

plt.show()
```



In [36]:

```
#Does RM experience have a relationship with CASA Lead conversion?
```

In [37]:

```
rm_performance = leads.groupby("rm_id").agg(
    total_leads=("lead_id", "count"),
    converted_leads=("converted", "sum"),
    conversion_rate=("converted", "mean")
)

rm_performance["conversion_rate"] = (
    rm_performance["conversion_rate"] * 100
)

rm_analysis = rm_performance.merge(
    rm[["rm_id", "experience_years", "monthly_target"]],
    on="rm_id",
    how="left"
)

rm_analysis = rm_analysis.sort_values(
    "conversion_rate",
    ascending=False
)
```

```
)  
  
print(rm_analysis)
```

	rm_id	total_leads	converted_leads	conversion_rate	experience_years	\
12	RM013	41	32	78.048780	9	
6	RM007	31	24	77.419355	6	
11	RM012	41	31	75.609756	7	
4	RM005	24	17	70.833333	7	
29	RM030	23	16	69.565217	12	
5	RM006	36	25	69.444444	6	
10	RM011	38	26	68.421053	2	
19	RM020	36	24	66.666667	4	
15	RM016	41	27	65.853659	14	
1	RM002	32	21	65.625000	3	
3	RM004	29	19	65.517241	2	
22	RM023	20	13	65.000000	14	
0	RM001	31	20	64.516129	6	
23	RM024	39	25	64.102564	7	
20	RM021	36	23	63.888889	5	
9	RM010	33	21	63.636364	8	
2	RM003	22	14	63.636364	13	
26	RM027	41	26	63.414634	3	
7	RM008	35	22	62.857143	5	
21	RM022	36	22	61.111111	14	
24	RM025	36	22	61.111111	8	
18	RM019	34	20	58.823529	1	
16	RM017	31	18	58.064516	10	
17	RM018	36	20	55.555556	12	
14	RM015	27	15	55.555556	10	
25	RM026	36	20	55.555556	14	
27	RM028	42	22	52.380952	6	
8	RM009	34	16	47.058824	1	
28	RM029	23	10	43.478261	11	
13	RM014	36	15	41.666667	14	

	monthly_target
12	38
6	58
11	25
4	26
29	79
5	74
10	79
19	69
15	43
1	33
3	25
22	26
0	37
23	37
20	25
9	54
2	41
26	43
7	71
21	34
24	60
18	62
16	32
17	24
14	76
25	43
27	33
8	66
28	37
13	60

Experience does NOT appear to automatically mean higher conversion.

For example:

RM014 → 14 years, but only 41.67% RM026 → 14 years, 55.56% RM013 → 9 years, 78.05% RM007 → 6 years, 77.42% RM009 → 1 year, 47.06%

So we cannot say "more experienced RMs convert more leads."

7)Sales data exploration

Chunk 7 — Sales Target Exploration

Objective: Understand target vs actual sales performance and identify whether sales achievement varies across observations.

currently compare:

Target vs Achieved

This will later support the business question:

Are branches achieving their targets?

```
In [38]: sales.head(20)
```

Out[38]:

	month	branch_id	target	achieved	variance
0	2026-01	B001	679	223	-456
1	2026-02	B001	679	269	-410
2	2026-03	B001	679	93	-586
3	2026-01	B002	414	345	-69
4	2026-02	B002	414	324	-90
5	2026-03	B002	414	366	-48
6	2026-01	B003	352	88	-264
7	2026-02	B003	352	329	-23
8	2026-03	B003	352	213	-139
9	2026-01	B004	516	81	-435
10	2026-02	B004	516	158	-358
11	2026-03	B004	516	252	-264
12	2026-01	B005	347	375	28
13	2026-02	B005	347	108	-239
14	2026-03	B005	347	386	39
15	2026-01	B006	558	164	-394
16	2026-02	B006	558	130	-428
17	2026-03	B006	558	440	-118
18	2026-01	B007	401	200	-201
19	2026-02	B007	401	99	-302

```
In [39]: sales.describe()
```

Out[39]:

	target	achieved	variance
count	30.000000	30.000000	30.000000
mean	500.000000	218.933333	-281.066667
std	118.005844	101.938093	174.255052
min	347.000000	81.000000	-586.000000
25%	401.000000	131.250000	-408.500000
50%	516.000000	206.500000	-281.500000
75%	558.000000	284.000000	-154.500000
max	688.000000	440.000000	39.000000

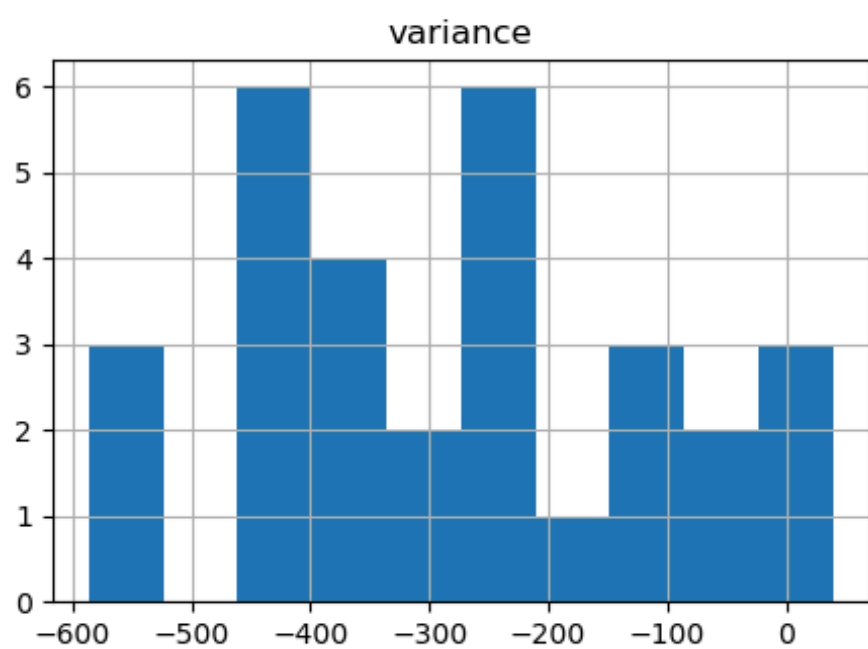
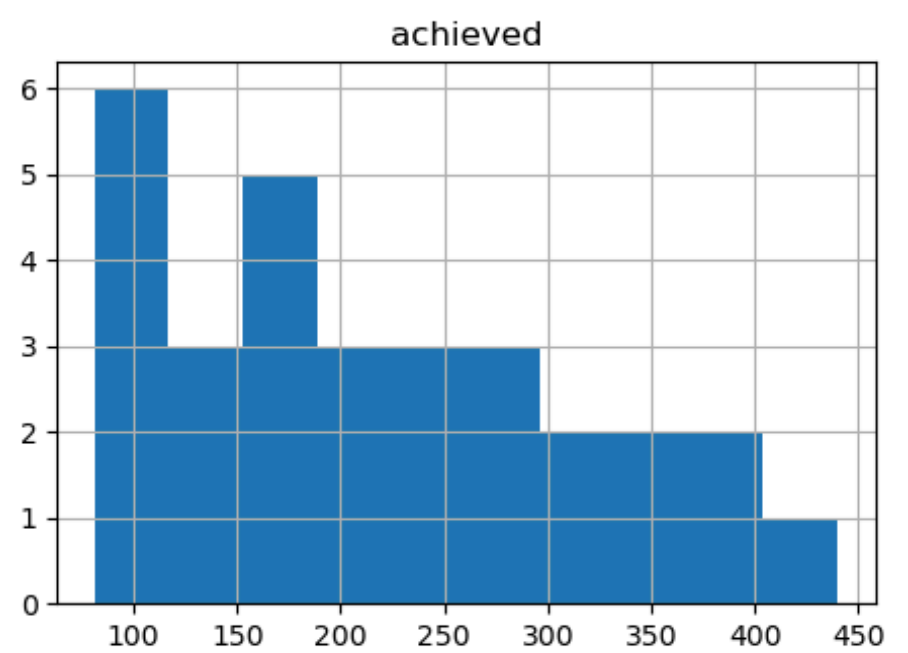
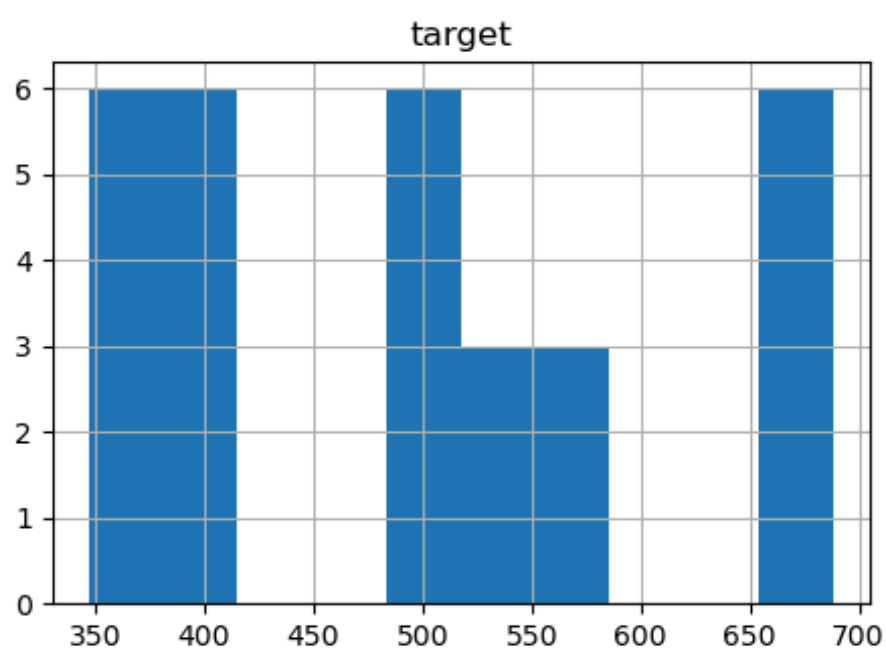
```
In [40]: sales.corr(numeric_only = True)
```

Out[40]:

	target	achieved	variance
target	1.000000	-0.251389	-0.824263
achieved	-0.251389	1.000000	0.755235
variance	-0.824263	0.755235	1.000000

```
In [41]: import matplotlib.pyplot as plt
sales.hist(figsize = (12,8), bins=10)
plt.tight_layout
plt.show
```

Out[41]: <function matplotlib.pyplot.show(close=None, block=None)>

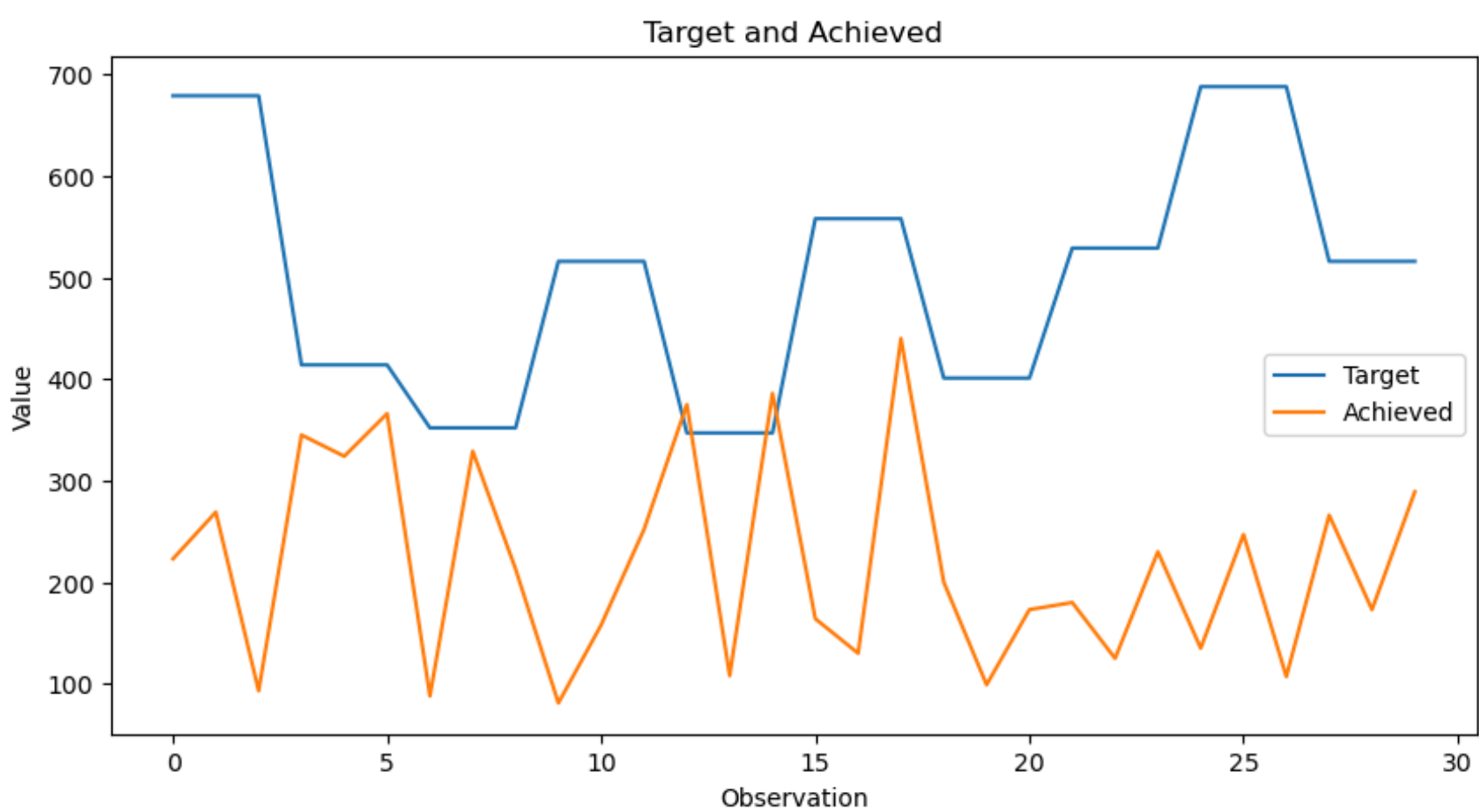


```
In [42]: plt.figure(figsize=(10,5))

plt.plot(sales["target"], label="Target")
plt.plot(sales["achieved"], label="Achieved")

plt.title("Target and Achieved")
plt.xlabel("Observation")
plt.ylabel("Value")
plt.legend()

plt.show()
```



8)Campaign data exploration

Chunk 8 — Campaign Exploration

Objective: Understand campaign activity and response levels across campaign types.

We analyze:

This can help determine which campaign types generate stronger customer responses.

```
In [43]: campaign.head(20)
```

Out[43]:

	lead_id	campaign_type	contacts	response
0	L00001	Call	3	Ignored
1	L00002	Call	4	Responded
2	L00003	Email	1	Ignored
3	L00004	Email	3	Ignored
4	L00005	SMS	3	Responded
5	L00006	Email	1	Responded
6	L00007	Call	5	Responded
7	L00008	SMS	5	Ignored
8	L00009	Email	4	Responded
9	L00010	SMS	1	Ignored
10	L00011	Email	1	Responded
11	L00012	Email	5	Responded
12	L00013	Call	1	Responded
13	L00014	Call	5	Responded
14	L00015	SMS	4	Ignored
15	L00016	Call	3	Responded
16	L00017	Email	5	Ignored
17	L00018	Email	1	Ignored
18	L00019	Email	2	Ignored
19	L00020	Email	4	Ignored

```
In [44]: campaign.isnull().sum()
```

Out[44]:

```
lead_id      0
campaign_type 0
contacts     0
response     0
dtype: int64
```

```
In [45]: campaign.columns
```

Out[45]:

```
Index(['lead_id', 'campaign_type', 'contacts', 'response'], dtype='object')
```

```
In [46]: campaign.groupby("campaign_type")["contacts"].sum()
```

Out[46]:

```
campaign_type
Call      963
Email    1025
SMS      1028
Name: contacts, dtype: int64
```

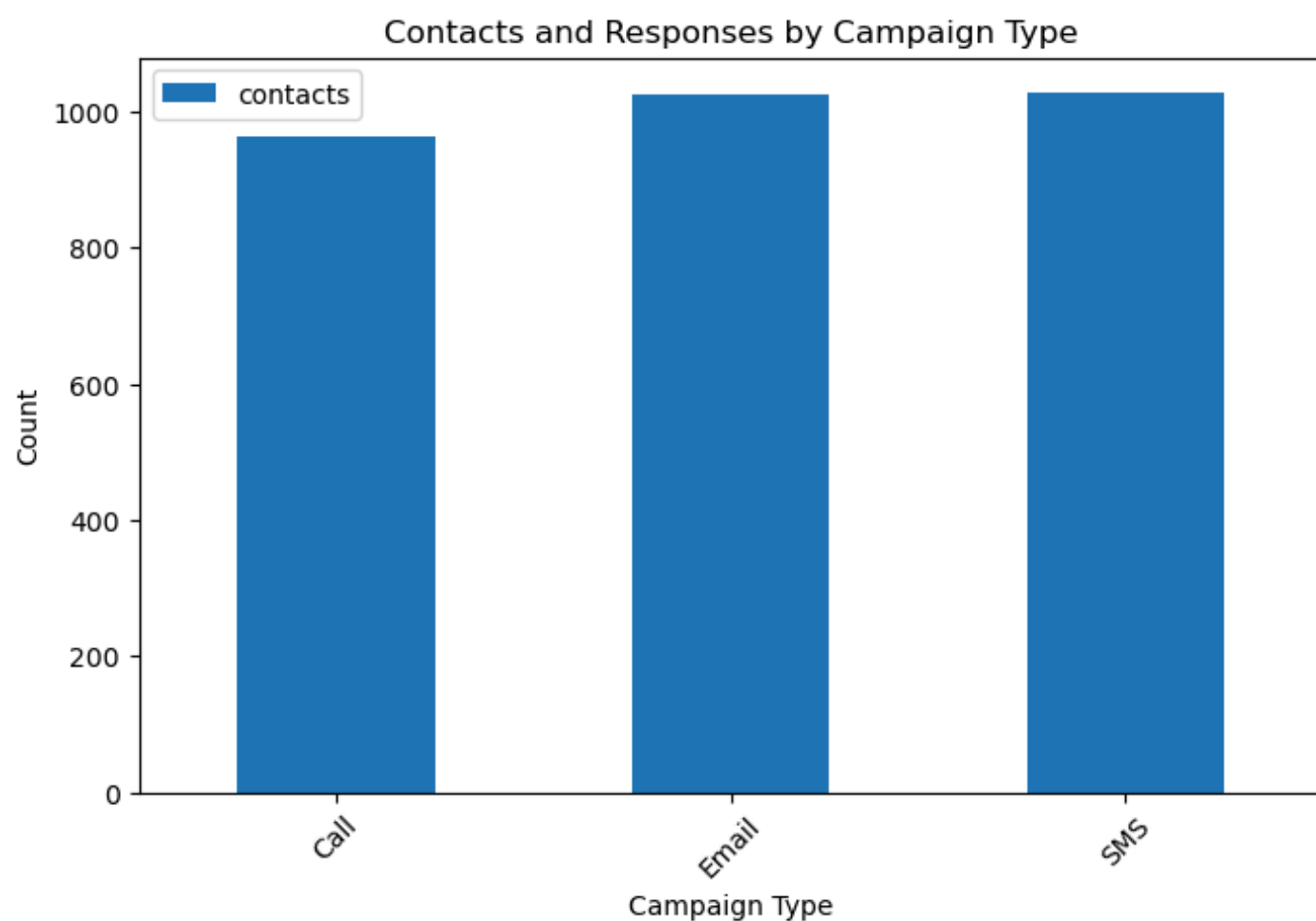
```
In [47]: import matplotlib.pyplot as plt

summary = campaign.groupby("campaign_type")["contacts", "response"].sum()

summary.plot(kind="bar", figsize=(8,5))

plt.title("Contacts and Responses by Campaign Type")
plt.xlabel("Campaign Type")
plt.ylabel("Count")
plt.xticks(rotation=45)

plt.show()
```



EDA Key Findings 1)Customer Data: Customer data was explored for distributions, missing values, duplicates, and potential outliers to ensure data readiness.

2)Branch Data: Branches have different monthly targets, but actual branch performance requires SQL-based integration with other tables.

3)Lead Conversion: Overall lead conversion was analyzed to understand the balance between converted and non-converted leads.

4)Lead Source: Referral leads had the highest conversion rate (65.13%), while Walk-in leads had the lowest (59.16%).

5)RM Performance: RM conversion rates varied considerably, from 41.67% (RM014) to 78.05% (RM013).

6)RM Experience: Higher RM experience did not consistently correspond to higher conversion rates.

7)CASA Accounts: Account data was explored for structure, descriptive statistics, missing values, and duplicate records.

8)Transactions: Transaction data was explored to identify customer activity patterns and potential variables related to CASA conversion.

9)Sales Performance: Target and achieved sales were compared, providing a basis for evaluating target achievement.

10)Campaign Performance: Campaign contacts and responses were compared across campaign types to understand campaign effectiveness.

11)Overall EDA Insight: Lead source and RM performance show noticeable differences in conversion, while customer activity and branch performance require further statistical and SQL-based analysis.

Next Step: Statistical testing will determine whether the observed relationships are statistically significant before proceeding to feature engineering and ML.

In []:

In []: